Lab 1: Occurrence Data, Niche Reasoning, and Reproducible Workflows

Today you will create the beginning of a reproducible workflow and decide what an occurrence dataset can and cannot support as ecological evidence.

What you need

- R, RStudio, GitHub Desktop, and a GitHub account.
- The unzipped `2026_R_Git_Niche_Reasoning` folder. Keep the file in `data/` unchanged.
- Your own computer, repository, code, written reasoning, and commits.

Part A

1. **Open the folder as an RStudio Project.** In RStudio select File → New Project → Existing Directory, then select the unzipped starter folder. This makes the project folder your working directory.
2. **Open the script.** In the Files pane open `analysis/01_occurrence_audit.R`. Save it before you change it.
3. **Run one line at a time.** Place the cursor on a line and use Run or Cmd/Ctrl + Enter. Do not type results by hand; keep the code that generated them.
4. **Confirm where R is working.** Run `getwd()` and `list.files()`. The output should show the starter folder and its `analysis`, `data`, and `output` folders.
5. **Read the data.** Run the guided block below. If it fails, stop and diagnose the working directory before changing the file path.
6. **Create your repository.** In GitHub Desktop choose File → Add Local Repository and select the starter folder. If prompted, create a repository there; keep it private. Commit with the message `Initialize occurrence-data lab project`, then Publish/Push.

```
getwd()
list.files()
occ_raw <- read.csv("data/crotalus_triseriatus_gbif_raw.csv", stringsAsFactors =
FALSE)
dim(occ_raw)
names(occ_raw)
head(occ_raw[, c("species", "decimalLongitude", "decimalLatitude",
"basisOfRecord")])
```

Why use a project rather than a hard-coded path? A project lets someone else open the same folder and run your script without rewriting paths on their computer.

Part B

Complete the problems in `analysis/01_occurrence_audit.R`. You may use R help (`?function_name`) and the function list at the end of this handout. The exact retained-record count may differ across students; your decisions must be visible, coded, and justified.

Problem 1: Audit the raw records

- Use code to report the number of rows, column names, data types, missing coordinates, and the frequencies of `basisOfRecord`.
- Inspect coordinate uncertainty, duplicated coordinates or records, event dates, and the GBIF `issue` field. What could each tell you about how these records were created?
- Write a short code comment identifying one feature you would investigate before treating records as analysis-ready.

Problem 2: Make two quality-control decisions

Write two named logical rules in the script. One rule must check coordinates. The other must consider duplicate records, sampling/provenance, dates, uncertainty, or another clearly stated issue. For each rule, state the spatial or temporal scale at which it makes sense and one valid record it might inadvertently exclude.

```
# Give your rules meaningful names. Replace these placeholders with your code.
keep_coordinates <- ...
keep_second_rule <- ...
keep <- keep_coordinates & keep_second_rule
occ_retained <- occ_raw[keep, ]

# Report how many records each decision changes.
sum(keep_coordinates)
sum(keep_second_rule)
sum(keep)
```

Problem 3: Make every decision visible

Create and save `output/occurrence_audit_map.png`. The raw records and retained/excluded records must be visually distinguishable. Include a title, axis labels, and a legend. A clean map is a diagnostic plot of observations, not a niche estimate or a distribution model.

```
png("output/occurrence_audit_map.png", width = 1600, height = 1200, res = 180)
plot(occ_raw$decimalLongitude, occ_raw$decimalLatitude,
     pch = 16, col = "grey70", xlab = "Longitude", ylab = "Latitude",
     main = "Crotalus triseriatus: raw and retained occurrence records")
points(occ_retained$decimalLongitude, occ_retained$decimalLatitude,
       pch = 16, col = "black")
legend("topleft", legend = c("Raw", "Retained"),
       pch = 16, col = c("grey70", "black"), bty = "n")
dev.off()
```

Problem 4: Return to the niche claim

Draft these answers in `README.md`; develop them into your assignment submission.

- What does your retained point cloud represent? What does it not represent?
- Which two environmental predictors would you obtain next? Explain why each is biologically relevant and give an appropriate spatial and temporal resolution with brief justification.
- What movement, dispersal-history, or geographic-barrier process might help explain why a species is not observed in every environmentally favorable place?

Before you leave class

- Save the script and commit what you completed, for example: `Audit raw occurrence records and draft QC rules`.
- Push the repository. If the push cannot finish in class, make a note of the exact step that failed and resolve it before the Friday submission.

Technical troubleshooting

- **`read.csv()` cannot find the file.** Run `getwd()` and `list.files()`. Reopen the starter folder as an RStudio Project before changing a path.
- **A command produces an error.** Read the first error line, rerun the immediately preceding object-creation line, and inspect that object with `str()`, `dim()` or `head()`.
- **Your map is blank or looks implausible.** Check for missing or reversed latitude/longitude values, then inspect the coordinate ranges before changing your quality-control rule.
- **GitHub Desktop will not publish.** Save the script, check that you are signed in, and look for the exact message. You may finish the push before Friday, but do not replace the repository with one final bulk upload.

Help without a solution

Useful base-R functions: `str()`, `summary()`, `names()`, `nrow()`, `is.na()`, `table()`, `duplicated()`, `paste()`, `round()`, `plot()`, `points()`, `legend()`, `png()`, and `dev.off()`. Start with `?function_name`, test a small piece of code, then explain what the result means.

Academic integrity: the syllabus permits AI tools for learning and troubleshooting, but the submitted code, reasoning, and interpretation must be yours and understandable to you. Briefly acknowledge any AI use in a comment or README note.

Reference

Peterson, A. T., Soberón, J., Pearson, R. G., Anderson, R. P., Martínez-Meyer, E., Nakamura, M., & Araújo, M. B. (2011). Ecological Niches and Geographic Distributions. Princeton University Press. Chapters 1–2.

Assignment 1 — submit Friday by midnight

Submit your individual private GitHub repository link to the canvas assignment. The repository must contain the following items.

- **Reproducible project and Git.** The project opens cleanly; raw data are preserved; the script runs; and commits describe the status of the project.
- **Audit and quality control.** Exploration of the data and two named rules are coded, quantified, ecologically justified, and cautious.
- **Diagnostic map.** Raw and retained/excluded records are distinguishable in a labelled, readable figure.
- **Niche-theory reasoning.** The README clearly distinguishes occurrences, environments, niche, and geographic distribution.
- **Next-step design.** Biologically relevant predictors, scale, and limiting processes are stated.

Your 200–300 word README should include: (1) data summary, (2) both quality-control decisions and their consequences, (3) what the map supports and does not support, and (4) the next environmental-data and the limiting processes influencing occurrences.